

WEEK 5 - Original Healthcare Dataset

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INTRODUCTION

The healthcare dataset was imported into Power BI and opened in Power Query for data preparation. The dataset contains patient details, medical conditions, admission information, hospital details, insurance provider, billing amount, medication and test results.

The dataset was inspected to understand the patient, admission and billing-related fields before performing data cleaning and transformation.

	Name	Age	Gender	Blood Type	Medical Condition	Date of Adm
1	Bobby JacksOn	30	Male	B-	Cancer	
2	LesLie TerRy	62	Male	A+	Obesity	
3	DaNnY sMith	76	Female	A-	Obesity	
4	andREW waTts	28	Female	O+	Diabetes	
5	adriENNE bEll	43	Female	AB+	Cancer	
6	EMILY JOHNsOn	36	Male	A+	Asthma	
7	edwArD EDWaRds	21	Female	AB-	Diabetes	
8	ChrisTinA MARTinez	20	Female	A+	Cancer	
9	JASmiNe aGullaR	82	Male	AB+	Asthma	
10	CHRISToPher BerG	58	Female	AB-	Cancer	
11	michELLE daniELs	72	Male	O+	Cancer	
12	aaRon MARtiNeZ	38	Female	A-	Hypertension	
13	connOR HANsEn	75	Female	A+	Diabetes	
14	rObErT bAuer	68	Female	AB+	Asthma	
15	bROOKE brady	44	Female	AB+	Cancer	
16	MS. nAtalie gAMble	46	Female	AB-	Obesity	
17	haley perkins	63	Female	A+	Arthritis	
18	mRS. jamiE cAMPBELl	38	Male	AB-	Obesity	
19	LuKE BuRgEss	34	Female	A-	Hypertension	
20	dANIEL schmidt	63	Male	B+	Asthma	
21	tiMOTHY burNs	67	Female	A-	Asthma	
22	CHRISToPHER BRIGHT	48	Male	B+	Asthma	
23	KathEDyn StauDe	68	Female	O+	Arthritis	

Cleaning Categorical Data

The categorical text columns were cleaned using the Trim and Clean transformations. This was done to remove unnecessary spaces and unwanted characters and to make the text values more consistent for analysis.

The Name column was also standardized using the Capitalize Each Word transformation to maintain consistent capitalization.

Transform Add Column View Tools Help

Transpose Data Type: Text Replace Values Unpivot Columns Reverse Rows Detect Data Type Fill Move Split Column Count Rows Rename Pivot Column Convert to List Merge Columns Format Extract Parse Statistics Standard Scientific Trigonometry Rounding Information Date Time Duration Run R script Run Python script

Any Column

Format

- lowercase
- UPPERCASE
- Capitalize Each Word
- Trim
- Clean
- Add Prefix
- Add Suffix

Table.TransformColumns(#"Sorted Rows",{{"Name", Text.Clean, type text}})

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission
1	Aaron Duncan	22	Male	AB-	Obesity	
2	Aaron Hicks	76	Female	A+	Arthritis	
3	Aaron Baldwin Jr.	20	Male	O-	Hypertension	
4	Aaron Hawkins	69	Female	B-	Diabetes	
5	Aaron Hart	18	Male	B-	Cancer	
6	Aaron Wilson	49	Female	O+	Obesity	
7	Aaron Foster	65	Male	AB+	Asthma	
8	Aaron Hopkins	26	Female	AB+	Asthma	
9	Aaron Greene	82	Male	AB+	Diabetes	
10	Aaron Oneal	69	Male	A-	Asthma	
11	Aaron Smith	46	Female	B-	Obesity	
12	Aaron Goodwin	59	Male	A-	Obesity	
13	Aaron Pope	41	Female	B-	Arthritis	
14	Aaron Smith	81	Male	A-	Cancer	
15	Aaron Smith	79	Male	A-	Cancer	
16	Aaron Bentley	18	Male	O-	Asthma	
17	Aaron Goodman	26	Female	AB-	Diabetes	
18	Aaron Rodriguez	63	Female	O-	Hypertension	
19	Aaron Richardson	20	Male	AB+	Cancer	
20	Aaron Romero	53	Male	AB-	Obesity	
21	Aaron Strickland	35	Female	B-	Cancer	
22	Aaron Anderson	50	Female	A+	Asthma	
23	Aaron Cook	37	Female	A+	Diabetes	

Queries [1]

healthcare_dataset

Table.TransformColumns(#"Capitalized Each Word",{{"Name", Text.Clean, type text}})

Query Settings

PROPERTIES

Name: healthcare_dataset

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Sorted Rows
- Capitalized Each Word
- Cleaned Text

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

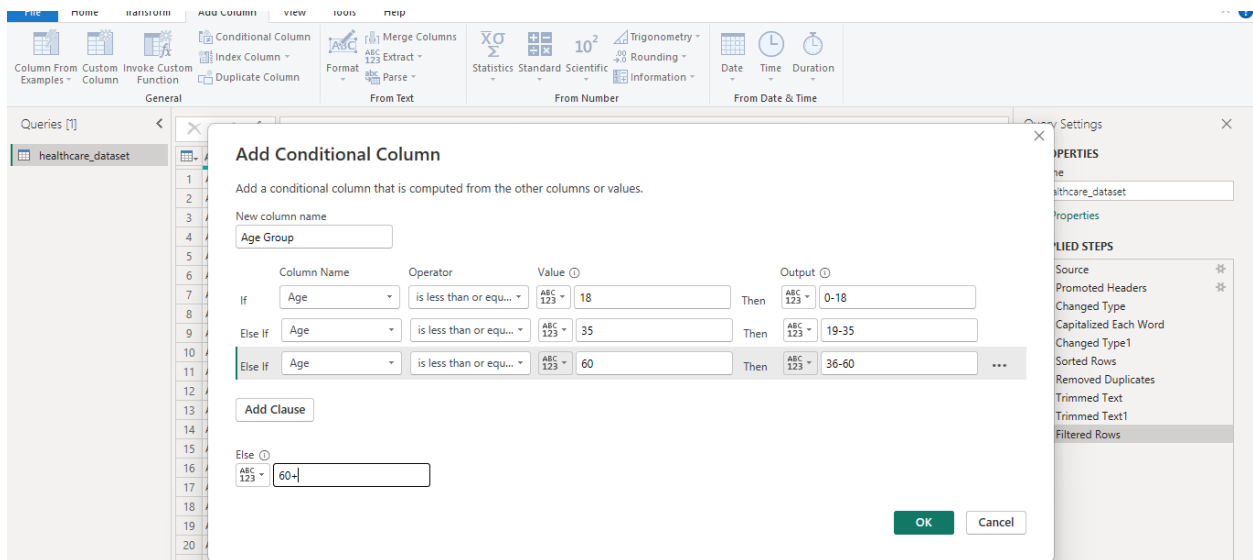
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Creating Age Group

A conditional column named Age Group was created based on the patient's age. The age values were divided into four groups: 0-18, 19-35, 36-60 and 60+. This helps in analysing patients based on different age ranges.

The Age Group column was created using a Conditional Column in Power Query.



Creating Dim Patient

A separate Dim_Patient table was created by keeping the patient-related columns. The table contains Name, Age, Gender, Blood Type and Age Group.

A Patient ID was added to uniquely identify the patient records and to connect the patient information with the Billing table.

A separate Billing table was created from the prepared healthcare data. The required billing and admission-related fields were retained for further analysis and modelling.

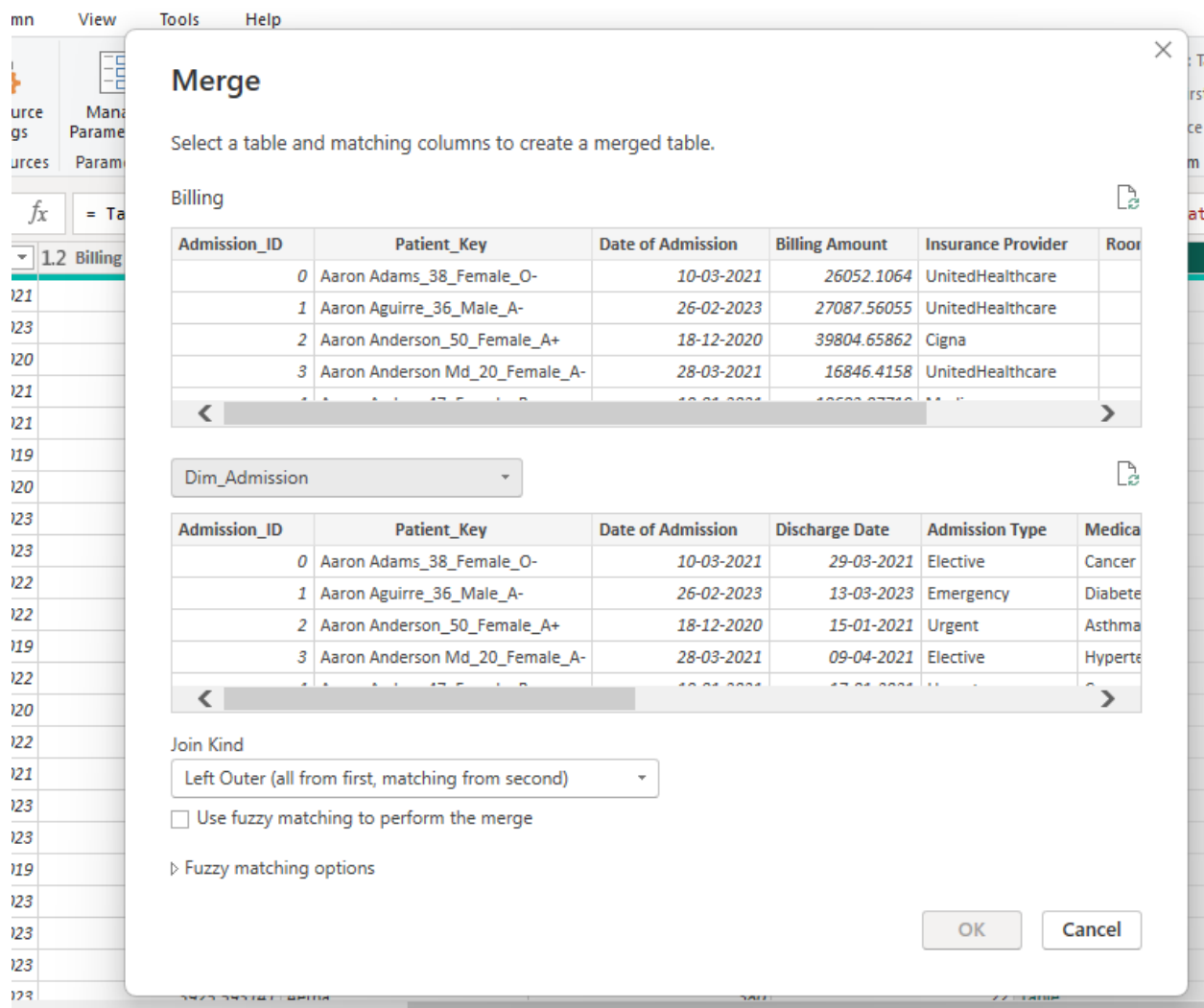
The Billing table contains important fields such as Date of Admission, Insurance Provider, Billing Amount and Discharge Date. Patient ID and Admission ID were also included to connect the Billing table with the dimension tables.

	Admission_ID	Patient_Key	Date of Admission	Billing Amount	Insurance Provider	Room
1	0	Aaron Adams_38_Female_O-	10-03-2021	26052.1064	UnitedHealthcare	
2		Aaron Aguirre_36_Male_A-	26-02-2023	27087.56055	UnitedHealthcare	
3		Aaron Anderson_50_Female_A+	18-12-2020	39804.65862	Cigna	
4		Aaron Anderson_Md_20_Female_A-	28-03-2021	16846.4158	UnitedHealthcare	
5		Aaron Archer_47_Female_B-	10-01-2021	10602.07719	Medicare	
6		Aaron Baker_73_Male_B+	18-06-2019	10135.88544	Medicare	
7		Aaron Baldwin Jr_20_Male_O-	10-10-2020	29740.9602	Medicare	
8		Aaron Barnes_85_Male_O-	30-03-2023	4783.444339	UnitedHealthcare	
9		Aaron Bautista_21_Male_B-	24-09-2023	49827.51977	UnitedHealthcare	
10		Aaron Bennett_76_Male_AB-	23-08-2022	12740.88681	Blue Cross	
11		Aaron Bentley_18_Male_O-	22-09-2022	5411.611014	Cigna	
12		Aaron Bradshaw_25_Female_O+	30-11-2019	16342.36373	Aetna	
13		Aaron Braun_55_Male_O+	02-12-2022	4436.082624	Blue Cross	
14		Aaron Brock_37_Male_AB+	18-01-2020	36225.56927	Aetna	
15		Aaron Brooks_26_Male_O+	10-08-2022	24756.15064	Blue Cross	
16		Aaron Bryant_77_Female_AB-	30-01-2021	32241.19907	Blue Cross	
17		Aaron Burns_67_Male_O-	09-06-2023	5495.133751	Aetna	
18		Aaron Byrd_47_Female_A+	24-12-2023	15448.88786	Aetna	
19		Aaron Campbell_42_Male_A-	21-08-2019	24972.32482	Cigna	
20		Aaron Cantu_18_Male_A+	18-06-2023	44959.48326	Blue Cross	

Merging with Dim Admission

The Billing table was merged with the Dim_Patient table using the matching patient information. A Left Outer join was used to retain all records from the Billing table.

After the merge, the Patient ID was expanded into the Billing table. This allows the billing records to be linked with the patient dimension.



Final Billing Table

The final Billing table was organized with the main fields required for analysis, including Date of Admission, Insurance Provider, Billing Amount, Discharge Date, Patient ID and Admission ID.

The table is prepared for use in the Power BI data model and dashboard analysis.

Query Settings

PROPERTIES

Name: Billing

APPLIED STEPS

Source: Removed Other Columns, Merged Queries, Expanded Dim_Patient

	Date of Admission	Billing Amount	Insurance Provider	Room Number	Dim_Patient.Patient ID
1	10-03-2021	26052.1064	UnitedHealthcare	363	0
2	26-02-2023	27087.56055	UnitedHealthcare	300	1
3	18-12-2020	39804.65862	Cigna	196	2
4	28-03-2021	16846.4158	UnitedHealthcare	249	3
5	10-01-2021	10602.07719	Medicare	108	4
6	18-06-2019	10135.88544	Medicare	234	5
7	10-10-2020	29740.9602	Medicare	104	6
8	30-03-2023	4783.444339	UnitedHealthcare	136	7
9	24-09-2023	49827.51977	UnitedHealthcare	465	8
10	23-08-2022	12740.88681	Blue Cross	312	9
11	22-09-2022	5411.611014	Cigna	500	10
12	30-11-2019	16342.36373	Aetna	255	11
13	02-12-2022	4436.082624	Blue Cross	481	12
14	18-01-2020	36225.56927	Aetna	291	13
15	10-08-2022	24756.15064	Blue Cross	337	14
16	30-01-2021	32241.19907	Blue Cross	111	15
17	09-06-2023	5495.133751	Aetna	156	16
18	24-12-2023	15448.88786	Aetna	211	17
19	21-08-2019	24972.32482	Cigna	366	18
20	18-06-2023	44959.48326	Blue Cross	275	19
21	14-02-2023	22312.85143	UnitedHealthcare	486	20
22	20-06-2023	29042.96183	Blue Cross	171	21
23	28-11-2023	8025.601141	Aetna	980	22

Final Data Model

The final data model contains Dim_Patient, Dim_Admission and Billing tables. The dimension tables are connected to the Billing table using Patient ID and Admission ID.

This structure separates patient and admission information from the main billing records and provides a structured model for healthcare dashboard analysis.

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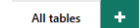
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Healthcare Analytics Dashboard

The healthcare dashboard was created using the cleaned and transformed data. It provides an overview of patient demographics, admission patterns, medical conditions and billing information.

The dashboard includes Total Patients, Total Billing Amount, Patients by Age Group, Patients by Gender, Patients by Admission Type, Patients by Medical Condition, Billing Amount by Medical Condition, Billing Amount by Insurance Provider and Patients by Test Results.



CONCLUSION

The healthcare dataset was successfully cleaned and transformed using Power Query in Power BI. The data was standardized, the Age Group column was created, and the data was organized into Dim_Patient, Dim_Admission and Billing tables. Patient ID and Admission ID were used to connect the tables and create a structured data model.

The final healthcare dashboard provides useful insights into patient demographics, admission types, medical conditions, test results and billing information. This structured model makes the healthcare data easier to analyse and supports effective visualization and reporting in Power BI.